Yanlong Yang

PhD Researcher — Computer Vision & Machine Learning

yanlong.yang@my.jcu.edu.au • +61 434 774 843 • Cairns, QLD, Australia
yyxr75.github.io • github.com/yyxr75 • Google Scholar • ORCID 0009-0000-8870-5993

RESEARCH INTERESTS

- Diffusion / generative models for inverse problems and computational imaging
- Multi-sensor fusion (radar, LiDAR, camera) for autonomous-driving perception
- 3D object detection, tracking, and point-cloud enhancement
- World models for perception

EDUCATION

PhD, Engineering — Computer Vision & Machine Learning Apr 2024 – Present

James Cook University, Cairns, Australia · Advisor: Dr. Tao Huang (Intelligent Computing & Communications Lab).

MEng by Research, Optical Engineering 2017 – 2020

Beijing Institute of Technology · Thesis: 3D human thermal-surface reconstruction (86.8 / 100).

BEng, Opto-Electronics Engineering 2013 – 2017

Tianjin University · Honours thesis on BST–silicon metamaterial THz spectra (86 / 100).

RESEARCH & PROFESSIONAL EXPERIENCE

Doctoral Researcher — James Cook University Apr 2024 – Present

Generative priors for radar super-resolution point-cloud generation; world models for perception. Contributed computer-vision models to a JCU × Queensland Government (DETSI) AI crocodile-detection system, trained on thousands of hours of video.

Algorithm Engineer — Vanjee Technology, Beijing 2020 – 2024

Radar–LiDAR perception for autonomous driving: 3D object detection and sensor fusion in adverse weather (basis of RaLiBEV). Led a 3-person team building a data closed-loop system (shipped v1.0); built a MIMO radar signal-processing toolkit.

Graduate Researcher — Beijing Institute of Technology 2018 – 2020

Designed the hardware, software, and mechanical system for human 3D thermal-surface reconstruction.

PUBLICATIONS

Peer-reviewed journals

★ RaLiBEV: Radar & LiDAR BEV Fusion for Anchor-Box-Free Object Detection 2025

Y. Yang*, J. Liu*, T. Huang, Q.-L. Han, G. Ma, B. Zhu. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(5):4130–4143.

Fusion Reconstruction Method for 3D Temperature Fields on the Human Body Surface 2022

Y. Yang, C. Xu. *Infrared Technology*.

Observation of Phase Transitions of Ba0.6Sr0.4TiO3–Silicon Hybrid Metamaterial by THz Spectra 2020

L. Wu, Y. Fu, N. Xu, Q. Sheng, X. Zhang, **Y. Yang**, et al. *ACS Applied Electronic Materials*.

Peer-reviewed conferences

SSP: Small-Object Detection from Sparse Roadside-LiDAR Point Clouds for V2I Systems 2023

H. Liang, **Y. Yang**, J. Li, Y. Deng. *ITS World Congress*.

Preprints & under review

★ Unsupervised Radar Point-Cloud Enhancement via Arbitrary LiDAR-Guided Diffusion Prior 2025

Y. Yang, J. Liu, G. Luo, H. Li, E. Ahn, M. Rahimi Azghadi, T. Huang. *arXiv:2505.09887*.

Self-Diffusion for Solving Inverse Problems 2025

G. Luo, S. Huang, **Y. Yang**. *arXiv:2510.21417*.

TECHNICAL SKILLS

Programming	Python (primary), C/C++, MATLAB · PyTorch, PCL, OpenGL, Qt
Tools	Git, LaTeX, COMSOL, Altium Designer, Visio, Linux
Domains	diffusion / generative modeling · inverse problems · sensor fusion · 3D reconstruction · autonomous-driving perception
Languages	Chinese (native), English (IELTS 6.5)

HONORS & AWARDS

2nd place (24th of 200 teams) — NXP Cup, China National University Smart-Car Competition 2017